

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
6.	(a)	(i)	I	the resistance in the parallel circuit is smaller (1) by 4 × (1) Accept converse answer Accept 'it' as parallel OR Total resistance in series circuit = 2 × R ₁ (1) Total resistance in parallel circuit = R ₁ /2 (1) OR The total parallel resistance is 4.5/4 (1) =1.125 (1)		2		2		3
			II	parallel is (4 ×) bigger than series OR Current is 8 A		1		1		
		(ii)		The voltmeter reading in the parallel circuit is more (1) by 2 × (than the series circuit) (1) OR Voltmeter reading is 9V in parallel (1) 4.5 V in series (1) OR The voltage is shared in series / not shared in parallel (1) So voltmeter reading is bigger in parallel (1)		2		2		1

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
		(iii)		The power {lost/dissipated} in R1 is greater in the parallel circuit (1) By 4 times (1)		2		2		2
	(b)	(i)		Current = $9/2.25$ (1) = 4 [A] (1) Power = 4 (ecf) $\times$ 9 (1) OR Power = $4^2 \times 2.25$ (1) = 36 [W] (1)	1 1	1 1		4	4	4
		(ii)		$\frac{1}{4}$ of 36W (ecf) (1) = 9W (1) OR $9/4.5 = 2$ (1) Power = $2 \times 4.5 = 9$ (1) OR $9/4.5 = 2$ (1) $2^2 \times 2.25 = 9$ [W](1)		2		2	2	2
				Question 6 total	2	11	0	13	6	10